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How a New Jersey
startup found an
electrifying way to slash
copper costs



Skyrocketing demand for copper promises to push prices to new heights. As the global economy transitions away from fossil fuels, it is going to need twice as much copper in the coming years than humanity has mined throughout all of its existence.

Still Bright, a New Jersey-based startup founded in 2022, thinks it has found a novel (and cleaner way) to slash those costs.

“The fact that we’ve already mined the easily mineable stuff, and the fact that we need many more mines to come into production every year — we’re talking like 60-plus mines — is it seems like an impossibility, like there’s no path to get there,” Randy Allen, co-founder and CEO of Still Bright, told TechCrunch.

But a large fraction of that demand could be met if companies can extract more copper from the ore they already mine.

Still Bright has developed a new way to extract copper, one that it says can recover nearly all the copper from typical ores without pre-processing steps that lose up to 20% of the metal. It’s effective enough that it could even be used on tailings, the discard piles that still contain smaller amounts of the metal that mines leave behind.

“Any copper that was lost as waste, we can actually process that and get the copper back,” Allen said.

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To boost production from single digits to hundreds of tons per year, Still Bright has raised an \$18.7 million seed round led by Material Impact and Breakthrough Energy Ventures, the company exclusively told TechCrunch. Azolla Ventures, Fortescue, Impact Science Ventures, and SOSV participated.

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Still Bright's technology allows it to extract copper without producing harmful pollution. Where most companies essentially burn away unwanted parts of the ore — releasing much of it into the atmosphere — Still Bright soaks copper-containing ores in a vanadium-based solution, which draws the metal out of the ore. When the vanadium solution is spent, the company's system uses electricity to regenerate it.

The core technology was inspired by a type of long-duration energy storage known as a "vanadium flow battery." In it, a vanadium-based solution that can

be stored in large tanks is charged and discharged by flowing it past a membrane.

“All of this is kind of happenstance. The technical inventor, the CTO of the company Jon [Vardner], he was working on two different projects,” Allen said, one on vanadium flow batteries and another on using vanadium to extract copper. “It’s connecting the dots. One person happened to be doing both.”

Still Bright’s modular system will be able to be installed at mines spanning a range of sizes.

Because the vanadium-based process works so quickly, the company’s equipment is much smaller than a typical refiner for the same amount of copper production. “The processing is on the order of minutes, up to an hour. That allows us to keep everything really small.”

The small size pays financial dividends, too. Allen said that Still Bright’s equipment is 70% to 90% cheaper than typical pyrometallurgical gear. Currently, the company’s process costs about the same to run as a typical refinery, but Allen expects that to change. “There’s a lot of opportunity for us to be cheaper,” he said.

Still Bright is planning to build a demonstration unit in 2027 or 2028 that’s capable of producing 500 tons of copper annually. It’s a big leap from the current pilot scale unit, which makes 2 tons per year. The ultimate commercial-scale system will produce 10,000 tons per year.

The clock is ticking, though. Still Bright would like to start refining copper in large enough quantities to benefit from any tariffs President Trump might impose on imports of the metal. If it does, it can use that revenue to develop and deploy commercial-scale units.

“We see ourselves as having a path to be among the cheapest copper producers,” Allen said.

This article has been updated to correct the name of one of the investors. Azolla Ventures, not Apollo Ventures, invested in Still Bright.

Topics: [Climate](#) [copper](#) [Startups](#) [TC](#) [Venture](#)



Tim De Chant

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Tim De Chant is a senior climate reporter at TechCrunch. He has written for a wide range of publications, including...

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Ceibo unearths \$30M Series B to extract more copper out of existing mines



When it comes time to fret about minerals critical to the energy transition, [lithium](#), cobalt and nickel may get all the attention, but another metal is keeping plenty of analysts up at night.

Though those other minerals are key ingredients for [batteries](#), copper is used across every sector that will be touched by electrification. Transmission wires, electric motors, inverters and battery packs all rely on the 29th element.

“There’s no energy transition, no decarbonization without some critical minerals,” [Ceibo](#) CEO Cristóbal Undurraga told TechCrunch+.

IMAGE CREDITS:

RIVERROCKPHOTOS / GETTY IMAGES

But less than a decade from now, demand for copper is expected to outstrip supply by 6.5 million metric tons, [according](#) to McKinsey. That portends a 25% shortfall.

One answer to that dilemma is to find more copper buried in the earth and dig it up. But that costs a lot, takes a ton of time and damages the environment. A better approach is to extract more copper out of the mines we have today.

There are several ways to do that — bacteria is an option; more on that later — but the one that helped Ceibo dig up a \$30 million Series B relies on tweaking a century-old process. The round was led by Energy Impact Partners with participation from existing investors Khosla Ventures and Aurus Ventures and new investors including CoTec

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A significant fraction of today’s copper is extracted by heap leaching. Rock containing copper ore is heaped into a pile and sprinkled with acid. As the acid trickles down through the pile, it strips some of the copper ore out of the rock. A basin at the base of the pile collects the acid-ore mixture, from which pure copper is extracted. It’s not the most environmentally friendly operation, but it uses less water and land than the alternative.



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One of heap leaching’s main drawbacks is that it only works on certain types of copper ore. Much of the copper ore dug from deep underground is of a type known as sulfides. Historically, heap leaching wouldn’t work on this type.

Ceibo has developed an additive to the acid that works on copper sulfides. As a result, mine operators who exhaust their easier-to-refine ores don’t have to build expensive facilities that would

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otherwise be needed to tackle the remaining sulfide ores. A few other companies are taking a similar approach.

That makes it sound easy, but it's not. The chemistry is challenging, sure, but the harder part is getting a foot in the door with mining companies, which are notoriously conservative and hesitant to try new techniques. "If you show up in a white coat, trust me, they're not gonna let you in the door," Undurraga said.

To get inside, Ceibo started by selling a dust suppressant and, eventually, a dust-suppression service. Mines, as you can imagine, kick up all kinds of dust, putting them in peril of angering the neighbors and violating their operating permits. The mining companies could spray water to keep it down, but many mines run in dry regions. So Ceibo's first offering promised to suppress 80% of a mine's dust and reduce water usage by 50% to 70%.

The dust suppression product "gives us some visibility," Undurraga said. "It also gives us some credibility, and it gives us some cash."

From there, the company started working on its leaching additive, which it's announcing today. It has also continued work on another, more intriguing technique involving bacteria. In fact, bacteria were always part of Ceibo's vision. "We started some years ago trying to understand if we could use this bacterial approach," he said. "We discovered that yes, you can. But it's going to take a lot of time to get scientifically."

The bacteria can be used in a range of applications, including on traditional heaps. But they can also be injected underground to digest the copper sulfates and be pumped back to the surface, copper in tow. Such an approach would minimize a mine's surface

disruption, though it would also pose risks to groundwater supplies.

Currently, the main technical challenge is wrangling bacteria in an open and uncontrolled environment. While bacteria live everywhere on Earth, coaxing them to do something extraordinary — like efficiently extracting copper deep below ground — usually involves some trade-offs. Bacteria may not thrive in those conditions, and those that do might not do the job quickly or effectively. Eventually, those trade-offs could be minimized, but it'll take time.

In the meantime, Ceibo is going to be using its Series B funding to transition from a dust-suppression company to one that helps mines recover more ore. So far, its phased go-to-market approach has helped build the business to a point where it has a reliable revenue stream. It was a deliberate decision, said Undurraga, who experienced the clean tech boom and crash in the early 2000s.

Cautious scaling isn't usually what startups prioritize, but in this case, it seems to be the right decision — one that's helped the company land a significant round with some solid investors. Next up: executing on the transition. If Ceibo can nail it, the looming copper shortage may not be so worrisome.

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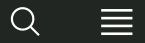
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SiTration uses silicon wafers to reclaim critical minerals from mining waste

Tim De Chant — 6:00 AM PDT · June 5, 2024

IMAGE CREDITS:

RIVERROCKPHOTOS / GETTY IMAGES

Brendan Smith didn't intend to be in the critical minerals business. He initially thought his high-tech filters would be good at desalination. But it was hard to beat existing systems, so he and his

Latest in Climate

adviser, MIT professor Jeffrey Grossman, quickly ruled it out. After a few years “where we sort of intentionally meandered,” they found themselves in a very different industry: mining.

Producing critical minerals is not a pretty business. Take copper, for example: The world uses around 25 million tons of copper per year, and more buildings, electronics, and electric vehicles mean that demand is [expected to grow](#) 24% over the next decade. Each ton of copper ore recovered leaves behind around [100–150 tons](#) of waste.

That’s a lot of waste, so much so that some mines have run out of room for tailings ponds, which hold the toxic chemicals and byproducts that remain after valuable minerals are removed.

“If you don’t have the land to build more of these ponds, you don’t have a lot of options,” Smith told TechCrunch. What’s more, because these ponds are open to the elements, rain and snow turn what could be a tractable problem into a never-ending one.

For many mine operators, dealing with waste is a costly part of the business. But Smith and Grossman’s company, SiTration, has a different proposal: use its equipment to treat the wastewater and harvest more minerals in the process.

“If you can do it for low cost, efficiently — which we can do — you can recoup some of the costs of remediation or even profit a little bit from that operation,” he said.

The key, Smith said, is silicon. SiTration’s filters are built on the same style of wafers that are used to make computer chips and solar panels, though using a cheaper grade of silicon. To turn the wafers into filters, Smith and Grossman tweaked an existing chemical treatment to etch minute pores in

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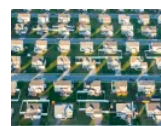


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Tim De Chant - Aug 1, 2025

them. The resulting material is more durable than existing membranes and, depending on what it's filtering, should last at least three years, Smith said.



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In a finished system, the company stacks a series of wafers atop one another and allows the liquids to flow through. For a larger mining site, the company would use about as much silicon as a medium-size solar farm, Smith said.

Because silicon is a semiconductor, SiTraction can send an electric current through the filters to make them even more effective for a range of minerals, including platinum, lithium, cobalt, and nickel.

Although SiTraction is starting with mining waste, including a pilot project [with Rio Tinto](#), it's also pitching its filtration systems to battery recyclers and metal refiners. In the case of metal refining, Smith said that his company's filters eliminate the need for heat while also reducing chemical use by up to 95%.

SiTration’s basic technology remains the same for each industry it sells to, Smith said. “We can ship out the same box to multiple partners and customers.”

To help build more boxes, the company has raised an \$11.8 million seed round led by 2150 with participation from Azolla Ventures, BHP Ventures, E14 Fund, Extantia, and Orion Industrial Ventures.

In the coming years, the company plans to deploy its filtration systems with partners in various industries. Depending on the size of the business using them, Smith said, “that could either constitute a very early demonstration for a very large volume mining partner or a pretty-close-to-commercial demonstration for a platinum group metal refiner.”

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BlueOak Raises \$35 Million For Its Factory That Mines Electronics

Jonathan Shieber — 4:39 AM PDT · June 10, 2014

[BlueOak Resources](#), a startup company backed by some of the biggest names in U.S. venture capital has raised \$35 million in project financing to mine precious metals in Arkansas.

[BlueOak](#) won't be strip-mining, open-pit mining, or moving mountains to get at precious metals. Instead the company intends to build mini-refineries that will extract the wealth of gold, silver, copper, and rare earth minerals buried deep in the bowels of the closets, drawers, and bags where the detritus that collects with each new release of the modern world's latest and greatest gizmos and gadgetry amasses.

In the U.S., [consumers alone dispose of some 3.2 million tons of e-waste](#) every year and more than 80% ends up in the trash, contributing to more than 70% of all toxic metals in U.S. landfills. Around the

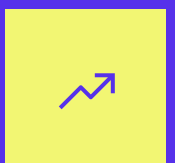
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world, only about 13% of e-waste undergoes any kind of recycling, according to a 2009 study from Advanced Technologies Materials Inc. Cities in China, like Guiyu, which is hailed as the international capital of e-waste, receives 4,000 tons of electronic garbage every hour.

For BlueOak, this trash is literally treasure. The company estimates that mining companies spend roughly \$12 billion per year searching for virgin ore deposits. While the U.S. throws out one ton of cell phones every 20 minutes, with each ton containing 70 times the amount of gold and silver found in virgin ore. For copper the number is even higher, with the equivalent of roughly one-third of global mining production thrown out in e-waste globally every year.

The company's first refinery, in Osceola, Ark., is backed by an initial commitment of \$35 million from the Arkansas Teachers' Retirement Fund, a consortium of undisclosed European and U.S. investors and the Arkansas Development Finance Authority. Production at the facility will begin in 2015, with the first factory able to process 15 million pounds of scrap per year.

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GOLD MINED USING BLUEOAK E-WASTE RECYCLING TECHNOLOGY.

BlueOak was launched in 2011 by Privahini Bradoo, a former executive at the emissions reduction and fuel production company LanzaTech New Zealand Ltd., and Bryce Goodman, a former analyst for Mitsubishi Corp. who specialized in rare earth minerals. The company raised an initial \$7.5 million from Kleiner Perkins Caufield & Byers and Boston-based Terawatt Ventures.

The idea for the company came from watching the development of the scrap-steel recycling business in the U.S. and the advent of mini-mill refining, according to Bradoo. “Fifty or sixty years ago the primary way you made steel was to dig a hole in the ground, extract ore, and smelt it. The mini-mill guys realized you could recover value from scrap steel

and set up facilities in some of the most rural parts of America.”



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Borrowing from that model, Bradoo envisions the possibility of setting up small refineries for e-waste across the U.S. It’s a vision that’s shared by some of the company’s investment partners, like the Heartland Renaissance Fund, the Los Angeles-based National New Markets Fund, and U.S. Bancorp Community Development Corp., which all provided tax credit financing to help BlueOak raise project finance for the facility.

“The New Markets Tax Credits were created with the vision of empowering and driving positive impact within rural communities in the U.S. and the BlueOak project allows us to do just that in Osceola,” said Sam Walls, the president of the Heartland Renaissance Fund, in a statement.

Photo via Flickr user [Sanguto](#)

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